The German artillery did just the same thing, as the following account from the British side shows:

> So regular were they [the Germans] in their choice of targets, times of shooting, and number of rounds fired, that … Colonel Jones … knew to a minute where the next shell would fall. His calculations were very accurate, and he was able to take what seemed to uninitiated Staff Officers big risks, knowing that the shelling would stop before he reached the place being shelled.

Axelrod remarks that such 'rituals of perfunctory and routine firing sent a double message. To the high command they conveyed aggression, but to the enemy they conveyed peace.'

The live-and-let-live system could have been worked out by verbal negotiation, by conscious strategists bargaining round a table. In fact it was not. It grew up as a series of local conventions, through people responding to one another's *behaviour*; the individual soldiers were probably hardly aware that the growing up was going on. This need not surprise us. The strategies in Axelrod's computer were definitely unconscious. It was their behaviour that defined them as nice or nasty, as forgiving or unforgiving, envious or the reverse. The programmers who designed them may have been any of these things, but that is irrelevant. A nice, forgiving, non-envious strategy could easily be programmed into a computer by a very nasty man. And vice versa. A strategy's niceness is recognized by its behaviour, not by its motives (for it has none) nor by the personality of its author (who has faded into the background by the time the program is running in the computer). A computer program can behave in a strategic manner, without being aware of its strategy or, indeed, of anything at all.

We are, of course, entirely familiar with the idea of unconscious strategists, or at least of strategists whose consciousness, if any, is irrelevant. Unconscious strategists abound in the pages of this book. Axelrod's programs are an exellent model for the way we, throughout the book, have been thinking of animals and plants, and indeed of genes. So it is natural to ask whether his optimistic conclusions—about the success of non-envious, forgiving niceness—also apply in the world of nature. The answer is yes, of course they do. The only conditions are that nature should sometimes set up games of Prisoner's Dilemma, that the shadow of the future should be long, and that the games should be nonzero sum games. These conditions are certainly met, all round the living kingdoms.